

Experiment #4

Digital Adder Circuits

OBJECTIVE

1. to understand how to construct half and full adders from basic combinational 74LS logic.
2. to be familiar with the functionality of the digital adder circuit. We will operate the 74LS83 4-bit parallel adder to get the feel of the adder operation.
3. Understanding how to design BCD adder/subtractor and to be familiar with its functionality.

BACKGROUND

We have recently studied the adder in class. We know that the basic “**half adder**” (HA) adds two bits to produce an arithmetic result and a possible carry. The basic diagram of the half-adder is given below:

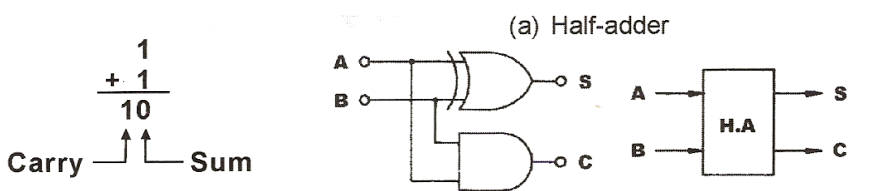


Fig. 1 Diagram of the half-adder

Symbol of HA

This circuit produces a 1 sum whenever A or B is 1, otherwise “Sum” is 0. However, when both A and B are 1 (and thereby Sum is 0), Carry is 1. This circuit produces a “complete” add function as long as there is no “carry-in,” i.e., as long as only two one-bit numbers are being added. If the numbers have more than one bit magnitude, however, all but the least significant bit (LSB) additions must have a carry-in, since there is the possibility of a carry being generated from addition of less significant bits in the number. In that case, the so called “**full adder**” (FA) must be used.

The full adder contains circuitry to accommodate a carry-in from addition of the two next least significant bits. Thus, addition of the two LSBs of two numbers can be made using half-adders, but full adders must be used to add the other bits of the two numbers.

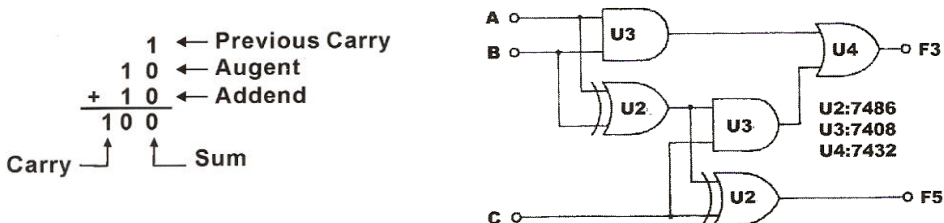


Fig. 2 Diagram of the Full-adder

“Full-Adder” (FA) can be build using two half adders as shown in the figure below

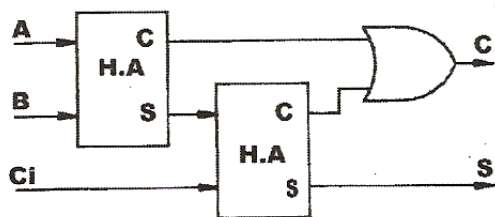
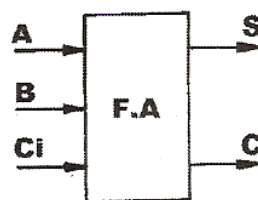


Fig. 3 Full-adder using two HA



Symbol of FA

To perform additions of numbers greater than 2-bits in length, the connection shown in Fig. 4, or “**Parallel Adder**” should be used to generate sums simultaneously. When FA1 adds A1 and B1, a sum S1 and a carry C1 is generated. C1 will be added to A2 and B2 by FA2, generating another sum S2 and another carry C2, and so on. The **7483** is a four bit binary parallel adder IC you can obtain its pin diagram from data sheets given online.

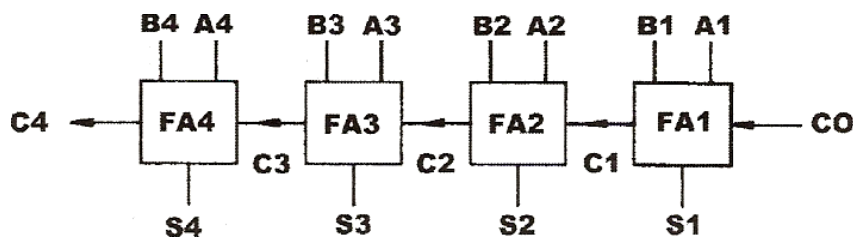


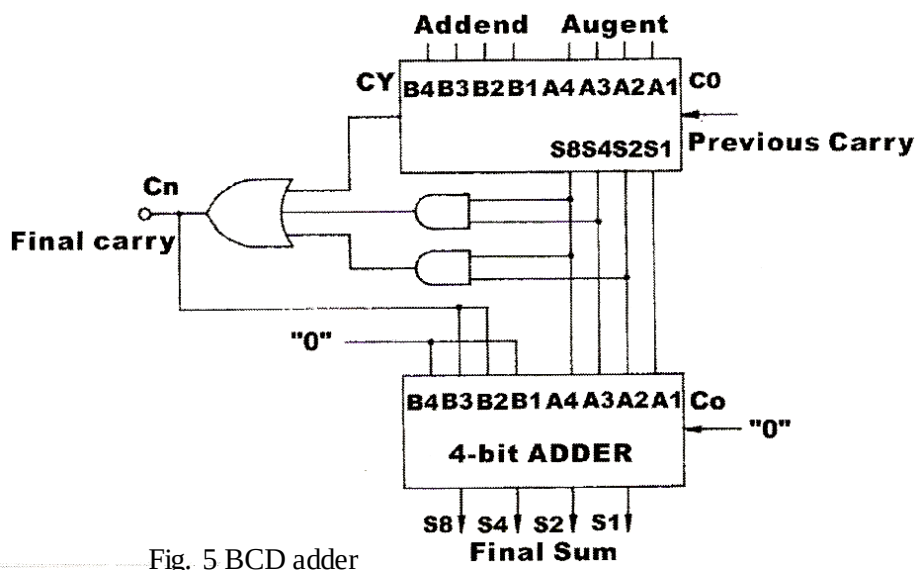
Fig. 4 four bit binary parallel adder

Binary adders can be converted into **BCD adders**. Since BCD has 4 bits with the largest number being 9; and the largest 4 bit binary number is equivalent to 15, there is a difference of 6 between the binary and the BCD adder. Under the following conditions 6 must be added when binary adders are used to add BCD codes:

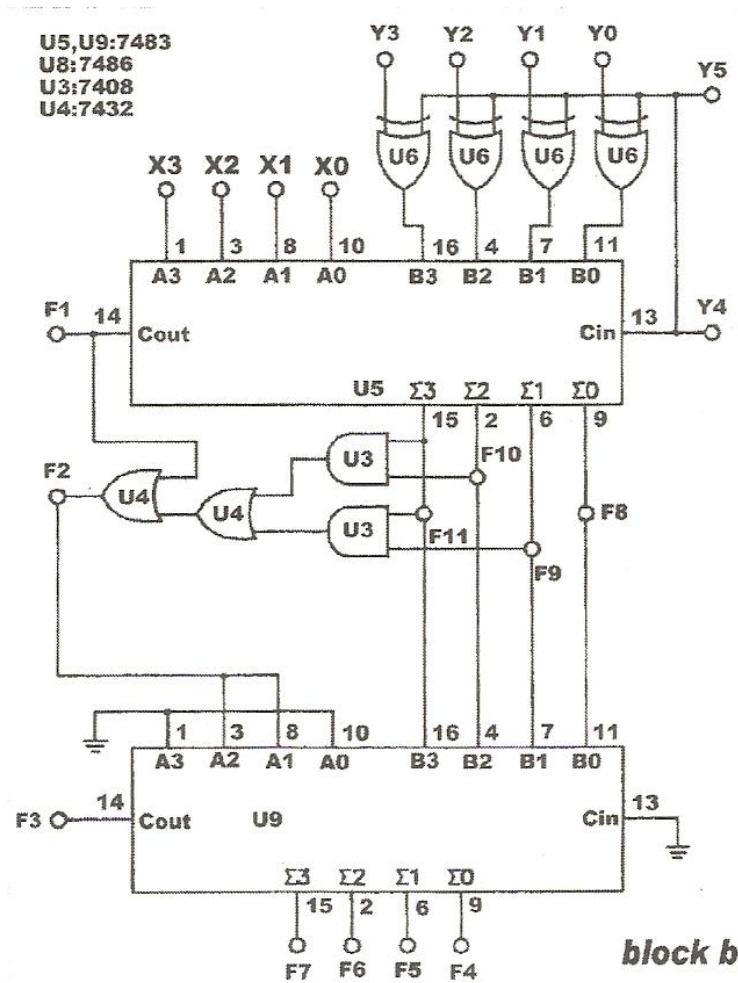
1. When there is any carry
2. When the sum is larger than 9

If the order of priority is S8, S4, S2, S1 and the sum is larger than 9 then $S8 \times S4 + S8 \times S2$. If any carry is involved, assuming the carry is CY, under this term, 6 must be added: $CY + S8 \times S4 + S8 \times S2$.

Fig.5 shows the circuit of BCD adder using two ICs of 7483 binary parallel adders, the function of the first adder U1 is to add the two bcd numbers X and Y, the function of other adder U2 is to add 6 to the result from U1 if the value of $Cn = CY + S8 \times S4 + S8 \times S2 = 1$ meaning sum is greater than 9 as mentioned above. Otherwise 0 is added.



The **BCD adder/subtractor** circuit is shown in fig.6 where the subtraction process is performed through out adding the 2's complement of the number to be subtracted. The circuit can work as adder when the input Y5 (the same as Y4) equals zero and as a subtractor when the input Y5 equals one.



PRELAB:

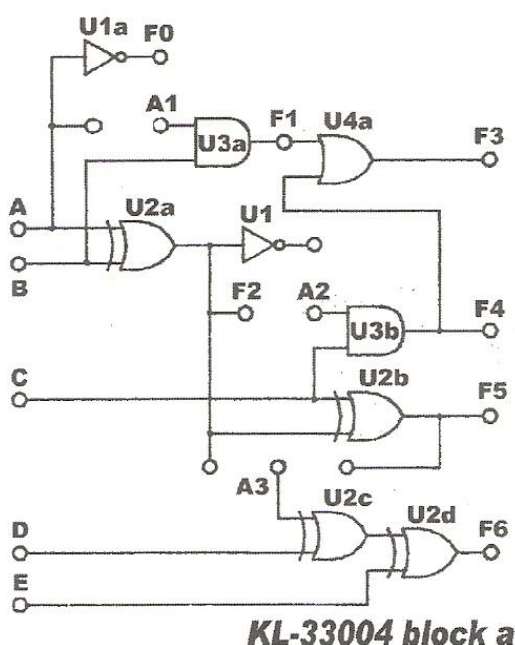
1. Draw pin diagram and truth table for HA circuit shown in fig.1.
2. Draw pin diagram and truth table for FA circuit shown in fig.2.
3. refer to your text book understand the circuits of BCD adder and subtractor.

EQUIPMENTS REQUIRED

KL-31001 trainer kit, lab module KL-33004

PROCEDURES:**Part I:**

- a) Construct the circuit of HA using module KL-33004 block a, connect inputs A and B to data switches and outputs F1 (carry) and F2 (sum) to LEDs, and do any other connections using clips. Record the truth table of the circuit and compare with that in your prelab.

**Part II:**

- a) Construct the circuit of FA using module KL-33004 block a, connect inputs A, B and C to data switches and outputs F3 (carry) and F5 (sum) to LEDs. and do any other connections using clips Record the truth table of the circuit and compare with that in your prelab.

Part III: Parallel-Adder Circuit with IC

- a) U5 on block b of module KL-33004 shown in fig.6 is used as a 4-bit parallel adder. Connect input Y5 to '0', so the XOR gates U6a-U6d, which are connected to Y0~Y3, will act as buffers.
- b) Connect inputs X0~X3 (addends), Y0~Y3 (augends) to DIP Switches DIP2.0~2.3 and DIP1.0~1.3 respectively. Connect F1, ΣO , ΣI , $\Sigma 2$, $\Sigma 3$ to

LEDs L1~L5. Follow input sequences in Table 1, record F1 and Z in hexadecimal numbers.

(X and Y can also be connected to the Thumbwheel Switches)

$$X = X_3 X_2 X_1 X_0$$

$$Y = Y_3 Y_2 Y_1 Y_0$$

$$\Sigma = Z_3 Z_2 Z_1 Z_0$$

Y	X	Σ	F1 (carry)
0	0		
0	1		
0	6		
0	9		
0	F		
1	3		
1	6		
1	8		
3	6		
4	8		
4	F		
8	7		
9	9		
A	B		
C	E		
F	F		

Table 1

Part IV: BCD adder Circuit

- The circuit on block b of module KL-33004 shown in fig.6 is used as a BCD adder.
- Connect input Y5 to '0', so the XOR gates U6a-U6d, which are connected to Y0~Y3, will act as buffers.
- Connect inputs X0~X3 (addends), Y0~Y3 (augends) to DIP Switches DIP2.0~2.3 and DIP1.0~1.3 respectively.
- Connect F8~F11 of U5 to the inputs of one of the 7-segment display (D0~D3 respectively). F8~F11 should also be connected to L1~L4. connect F1, F2 to logic indicators L5, L6.
- Connect outputs F4~F7 of U9 to the inputs of another 7-segment display (D0~D3 respectively). F4~F7 should also be connected to L7~L10 and F3, to logic indicators L11.

- f) F8~F11 are the sum of X0~X3 added to Y0~Y3 while F1 is the carry.
Follow the input sequences for X0~X3 and Y0~Y3 in table 2 and record the output states.
- g) Discuss your results.

INPUTS								OUTPUT (U5)					LAST (U9)					
X3	X2	X1	X0	Y3	Y2	Y1	Y0	F1	F11	F10	F9	F8	F2	F3	F7	F6	F5	F4
0	0	0	0	0	0	0	0											
0	0	0	1	0	0	1	1											
0	0	1	1	0	1	0	0											
0	0	1	0	0	0	1	0											
0	0	1	0	1	0	0	0											
0	0	1	1	0	1	1	0											
0	1	0	0	0	0	1	0											
0	1	0	0	0	1	0	1											
0	1	0	0	0	1	1	0											
0	1	0	1	0	1	1	0											
0	1	1	0	0	1	1	1											
0	1	1	1	1	0	0	0											
0	1	1	1	1	0	0	1											
1	0	0	0	1	0	0	1											
1	0	0	1	1	0	0	1											
1	0	1	0	1	0	1	0											
1	0	1	0	1	0	1	1											
1	0	1	0	1	1	0	0											
1	0	1	1	1	1	1	0											
1	1	1	1	1	1	1	1											

Table 2

Exercise:

- 1) Repeat part III, this time connect Y5 to logic 1 for the XOR gates to act as inverters and the circuit now is a subtractor. Fill in another table like table 1, discuss your results and verify that your circuit is working as expected.